11/03/23

(Please write your Exam Roll No.) *Exam Roll No. .....*

END TERM EXAMINATION

FIRST SEMESTER [B.TECH] MARCH 2023

Paper Code: ETPH-103 ***Subject: Applied Physics-I***

Time: 3 Hours ***Maximum Marks: 75***

Note: Attempt five questions in all including Q.No.1.which is compulsory. Select one questions from each unit. Assume missing data, if any.

Q1. Answer the following questions:- **[2.5x10=25]**

(a)Write the necessary conditions for obtaining interference fringes?

(b)What are the advantages of increasing the number of ruling in a grating?

(c)What is a half wave plate and how is it formed?

(d)Explain the term 'mode' related to optical fibre.

(e)What are the conditions of laser action?

(f) Find the speed of an electron having mass doubled its rest mass.

(g)State the postulates of Einsteins Special Theory of relativity.

(h)What are the advantages of cyclotron over linear accelerator?

(i)Find the frequency of rotation of a proton in a cyclotron whose magnetic fields is IT. Assume that the mass of the proton is $1.67 * 10^{-27}$ Kg?

(j)Differentiate between uniaxial and biaxial crystals.

UNIT-I

Q2. (a)With the help of a labelled ray diagram discuss the formation of Newton's rings by reflected light. Hence derive an expression for the diameter of nth dark ring. **[6]**

(b)What do you understand by Fresnel's Biprism and explain the formation of fringes by it. How do you determine the wavelength of monochromatic light? **[6.5]**

Q3. (a) Describe Fraunhofer diffraction due to single slit and deduce the position the position of maxima and minima. **[8]**

(b)Angle of second order diffraction is found to be 30° when a beam of light (λ=6000 A°) is made incident perpendicular on a plane transmission diffraction grating. Calculate number of lines per cm of the grating. **[4.5]**

UNIT-II

Q4. (a) Describe in brief the phenomenon of double refraction. Compare the properties of ordinary and extraordinary rays. **[7]**

(b) Define polarisation of light? What is the polarising angle for glass whose refractive index for light used in 1.6827 **[5.5]**

Q5. (a) How does the light propagate in an optical fiber. Explain numerical aperture and acceptance angle of an optical fiber. **[6]**

(b)What are Einstein's co-efficients? Obtain a relation between them. Also discuss the essential conditions for laser action. **[6.5]**

P.T.O.